SPHERE MISSION CONTROL

Thermal Insulation Laboratory

Standard Operating Procedure

Glass jar immersion test with an ice-water boundary

Document control	Value
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Purpose. This procedure enables four student teams to compare the transient cooling of identical, sealed 100 mL glass-jar capsules protected by bare, bubble-wrap, Mylar, and multilayer test configurations. It aligns the physical experiment with the current SPHERE student manual, teacher manual, task sheet, flashcards, and Mission Control model.

SAFETY - Only the instructor or another designated adult may prepare and dispense hot water. The maximum initial water temperature is 80.0 °C. Never use boiling water. Eye protection and heat-resistant gloves are required.

1 Scope and learning outcomes

This is an educational analog for comparing heat-transfer mechanisms and effective insulation performance. The glass jars, cotton, bubble wrap, and reflective film are not flight-qualified hardware, and an ice-water bath does not reproduce a space vacuum. The experiment instead provides a repeatable cold boundary for model comparison.

After the activity, students should be able to distinguish conduction, convection, and radiation; apply Newton's law of cooling; record a controlled temperature time series; calculate residuals and mean absolute error (MAE); and explain how leakage, probe placement, layer compression, and boundary drift affect repeatability.

2 Roles and stop work authority

Role	Responsibility
Instructor	Complete the local risk assessment; supervise hot-water handling; verify jar condition, probe-lid seals, ice-bath temperature, and safe separation of water from electronics.
Student teams	Assemble the assigned insulation configuration, record measurements exactly on schedule, report anomalies immediately, and keep the work area dry.
Timekeeper and data lead	Start the timer at immersion, announce 60-second intervals, enter values in Mission Control and the task sheet, and document late or missed readings.
All participants	Stop the test if a jar cracks or leaks, insulation becomes saturated, a probe moves, water reaches electronics, or any person is at risk.

3 Equipment and materials

Figure 1. Current SPHERE toolkit.

Qty.	Item	Requirement
4	100 mL wide-mouth glass jars with prepared probe lids	Heat-safe, undamaged, identical geometry. The lid opening supports the probe and approved classroom sealing method.
4	Waterproof digital probe thermometers	Stainless-steel probes; document instrument accuracy. Manuals use ±0.5 °C as the reference.
4	Ice-water bath containers	Stable and deep enough for the external water line to remain above the internal capsule water level.
1 set	Cotton, bubble wrap, reflective Mylar, tape or bands	Dry and applied consistently without crushing trapped-air layers.
1	Supervised hot-water source and measuring vessel	Instructor use only; prepare no hotter than 80.0 °C.
1+	Bath thermometer	Independent measurement of the actual boundary temperature Tenv.
-	PPE, towels, timer, worksheets	Required before the hot-water phase. Keep electronics at a separate dry station.

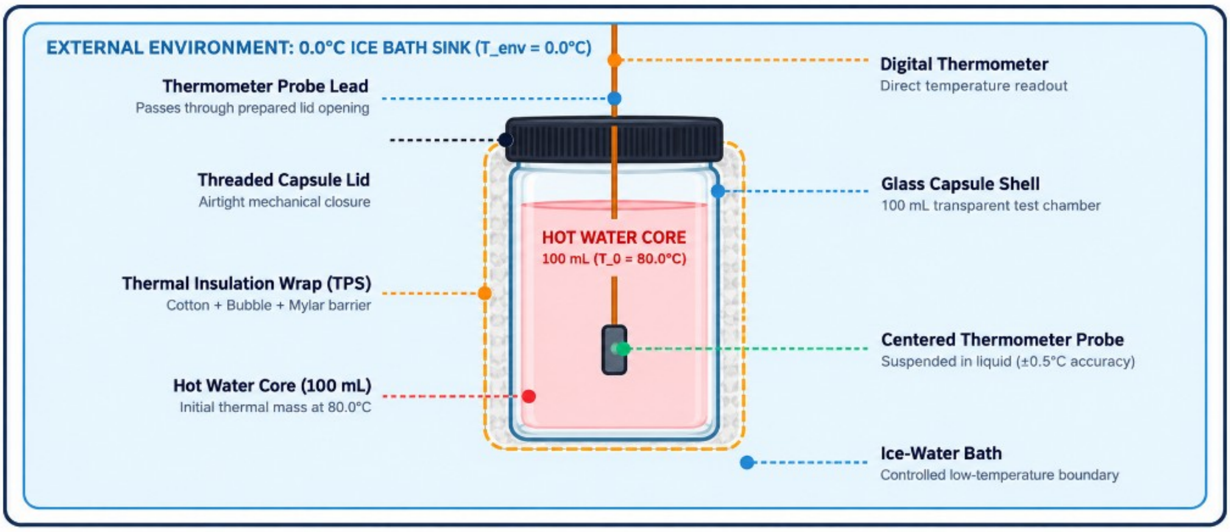


Figure 2. Glass-jar arrangement and centered probe placement.

4 Safety controls

SAFETY - Hot water can cause scalding. Do not exceed 80.0 ℃, do not use boiling water, and do not allow unsupervised hot-water handling.

- Inspect every glass jar and lid. Reject chipped, cracked, scratched, or poorly fitting components.
- Perform the seal check with cold water before hot filling. Use only the approved classroom lid-sealing method; do not improvise a pressure seal.
- The capsule is closed to resist leakage but is not a pressure vessel. Do not overfill, pressurize, or aim it toward a person.
- Wear eye protection and use heat-resistant gloves or suitable tongs for filling, closing, transferring, immersing, and removing a warm capsule.
- Keep the bath below waist height on a stable surface. Wipe spills immediately; keep displays, cables, computers, and power supplies dry.
- Only the intended waterproof metal probe may contact water. The display, connector, and lead junction remain dry.
- For hot-room operation, use 55-60 ℃ initial water or a room-temperature bath and enter the actual T0 and Tenv in Mission Control.

5 Pre flight preparation

Figure 3. Complete theory and safety review before handling hot water.

1. Confirm the risk assessment, PPE, dry electronics station, towels, and emergency response arrangements.
2. Fill each jar with cold water, fit the probe through its prepared lid, secure the opening using the approved classroom method, close the lid, and invert briefly over a sink or secondary container. Reject or reseal leaks.
3. Set the probe at the vertical and horizontal center of the 100 mL water volume. Mark or clamp the lead so depth cannot change.
4. Prepare a well-mixed ice-water slurry. Measure and record the actual bath temperature; do not assume exactly 0.0 ℃.
5. Assign configurations A-D. Use identical water mass, jar geometry, probe depth, bath depth, timing, and handling.

CONTROL - A fair comparison requires the same starting-temperature tolerance, water volume, probe position, immersion depth, bath condition, and 60-second interval.

6 Insulation configurations

The constants below are educational reference-model values, not certified material properties or guaranteed results. The symbol k is an effective system cooling constant, not thermal conductivity.

Configuration	k (min^{-1})	Construction	Model T(15)
A Bare	0.150	Uninsulated glass jar.	8.4 ℃
B Bubble wrap	0.040	Two even layers with intact cells; do not compress.	43.9 ℃
C Mylar	0.121	One reflective layer, shiny side outward; avoid tears and direct water paths.	13.0 ℃

Configuration	k (min^{-1})	Construction	Model T(15)
D Multilayer	0.015	Inner cotton, middle bubble wrap, outer Mylar; secure independently.	63.9 ℃

Model temperatures use T0 = 80.0 ℃ and Tenv = 0.0 ℃. Cotton-only is an optional fifth trial.

A Bare glass capsule

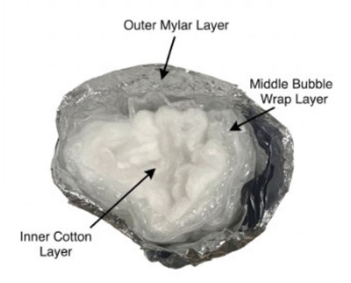
B Bubble wrap

C Mylar

Figure 4. Standard single-configuration assemblies.

6.1 Multilayer assembly sequence

1. Wrap the dry glass jar uniformly with cotton. Avoid thick folds and leave the lid functional.
2. Add bubble wrap with consistent overlap and intact cells. Do not crush the bubbles.
3. Add Mylar as the outer layer, shiny side outward. Limit water circulation into layers while keeping the display and lead junction dry.
4. Confirm the probe has not shifted and record layer order and approximate coverage.



Layer order

Completed assembly

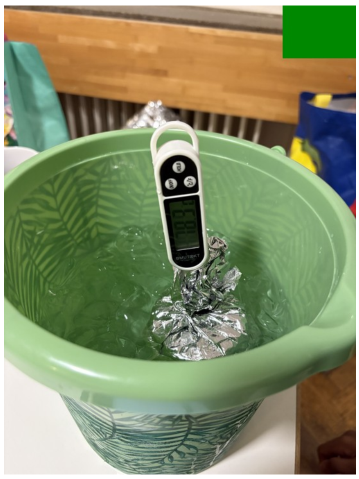
Figure 5. Educational multilayer assembly; not flight-qualified MLI.

7 Experimental procedure

Four teams should run in parallel. If one jar is reused, return it and the bath to the same measured initial conditions before each run.

Step	Action	Method and acceptance point
1	Configure Mission Control	Select the assigned model, enter measured T0 and Tenv, and run the prediction before collection.
2	Prepare the bath	Maintain a mixed ice-water slurry; record Tenv immediately before immersion and at the end.
3	Prepare the capsule	Instructor adds exactly 100 mL at no more than 80.0 °C and closes the prepared probe lid using the approved method.
4	Verify start	Confirm the internal reading is stable and within the agreed tolerance. Reposition only before the run.
5	Immerse and record T0	Use gloves or tongs. External bath level must exceed the internal water level. Start timing at immersion; keep display and lead junction dry.
6	Log 15 minutes	Record internal temperature at t = 1, 2, ..., 15 min. Do not move or squeeze the sample. Document late or missed readings.
7	Monitor controls	Watch seal, bath level and temperature, probe position, and wetting. Stop for leakage, glass damage, unstable supports, or electrical exposure.
8	End the run	At 15 minutes record capsule and bath temperatures; remove using gloves or tongs into secondary containment.
9	Check data	Confirm 16 observations from 0 through 15 minutes. Retain the Mission Control export and paper task sheet.
10	Cool and clean	Cool before unwrapping. Empty at a sink and dry jars, probes, insulation, bath, and work surface.

Measure actual bath boundary



Immerse to internal water line

Figure 6. Boundary verification and capsule immersion.

Figure 7. Keep the dry data station separate from the bath.

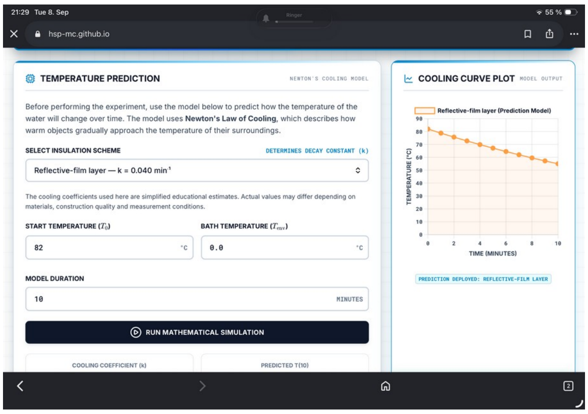
8 Data reduction and model comparison

Use the actual measured bath temperature as Tenv. The governing equations are:

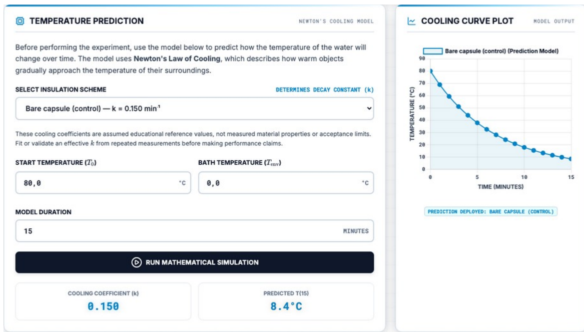
$$
\begin{aligned}
T_{\text{model}}(t) &= T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}, \\
r_i &= T_{\text{obs},i} - T_{\text{model},i}, \\
\text{MAE} &= \frac{1}{n}\sum_{i=1}^{n} |r_i|, \\
k_{\text{est}} &= -\frac{1}{t}\ln\left(\frac{T(t) - T_{\text{env}}}{T_0 - T_{\text{env}}}\right).
\end{aligned}
$$

Equation set 1. Newtonian cooling, residual, MAE, and estimated effective cooling constant.

Use n = 16 when all readings from t = 0 through 15 minutes are included. State a different n if a reading is invalid or missing. Report k in min^{-1} and MAE in ℃. Compare MAE with instrument accuracy and repeated-run variation; do not convert MAE into an invented percentage score.



Mission Control setup



Saved reference prediction

Figure 8. Configure and preserve the prediction before measurement.

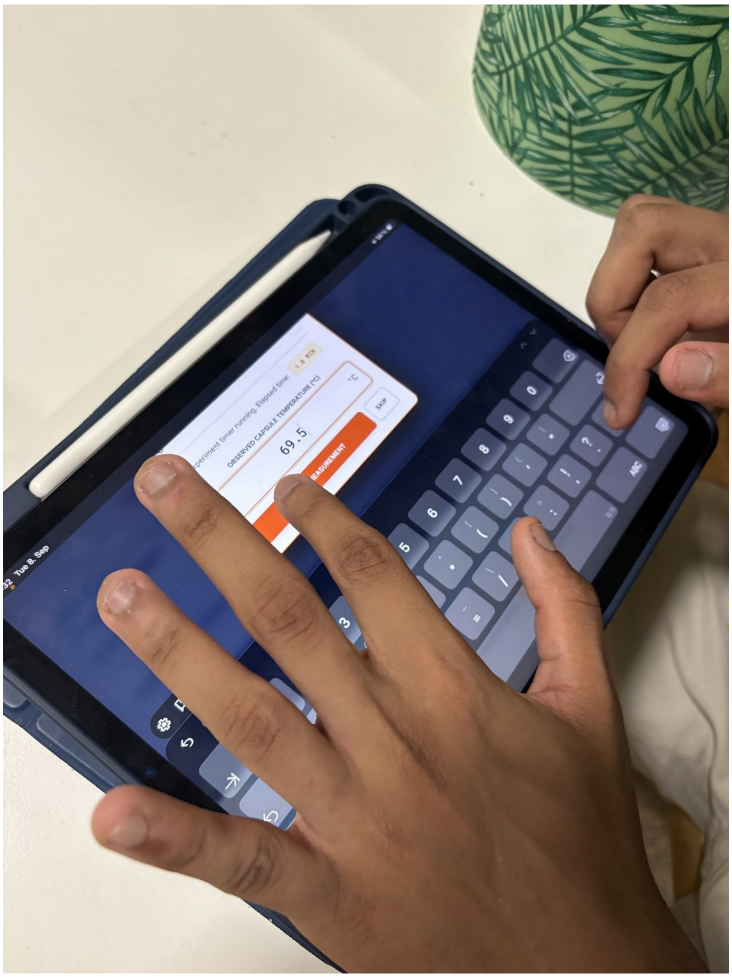


Figure 9. Manual temperature entry at the actual timestamp.

9 Quality controls and troubleshooting

Observation	Likely cause	Response
Sudden fall	Leakage, probe touching glass, sample moved, severe mixing.	Stop if leakage is suspected; mark invalid, inspect after cooling, and repeat safely.
Starts differ	Filling or transfer delay, inconsistent water source.	Repeat within tolerance or compare fitted k and normalized temperature excess.
Bath warms	Too little ice, poor mixing, high room temperature.	Measure actual temperature, update Tenv, and document drift.

Observation	Likely cause	Response
Mylar is weak	Direct liquid contact, tears, water entry, no spacer.	Treat as immersed system behavior, not universal material failure.
MLI varies	Different thickness, crushed cells, open top, overlap.	Photograph construction; rebuild to the standard sequence.
High MAE	Wrong model input, timing, probe shift, leakage, uncalibrated k.	Check inputs and timestamps, then discuss uncertainty and model limits.

10 Acceptance checklist

☐ Four undamaged 100 mL glass jars with prepared probe lids were used.

☐ Every lid passed a cold-water leak test before hot filling.

☐ Initial water temperature did not exceed 80.0 ℃ and actual T0 was recorded.

☐ Actual ice-bath temperature was measured and entered as Tenv.

☐ Probe depth, water volume, immersion depth, timing, and layer coverage were controlled.

☐ Every valid trial contains 16 time-temperature pairs from 0 through 15 minutes.

☐ Graphs include units, labels, a legend, observed data, and model data.

☐ Residuals and MAE use the stated number of valid observations.

☐ Conclusions distinguish system behavior, build quality, uncertainty, and assumptions.

☐ All equipment and surfaces were cooled, emptied, dried, and stored safely.

11 Document synchronization record

This document uses the same experiment definition as the current project materials: 100 mL glass jars with prepared probe lids; measured ice-water boundary; four standard configurations; a 15-minute run with observations from t = 0 through 15 minutes; Newtonian cooling; residuals and MAE; and the Mission Control reference constants.

Aligned source	Controlled content
Student Manual	Hardware, configurations, layer sequence, stages, equations, and MAE.
Teacher Manual	Safety limits, risk controls, lesson flow, constants, and instructor criteria.
Student Task Sheet	Sixteen-point log, graph, residual and MAE fields, and ten inquiries.
Flashcards	Heat-transfer, effective k, boundary condition, and multilayer terminology.
Mission Control web app	Configuration names, constants, timer, manual entry, and MAE output.

Limit of use. This activity demonstrates comparative thermal behavior under classroom conditions. It does not certify any material or assembly for aerospace, pressure-vessel, fire-protection, or personal-protective use.